Rithwik Mohanty

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EDUCATION

University of Connecticut · Honors Program Expected Dec 2027
B.S. in Mathematics and Statistics, Minor in Computer Science
Relevant Coursework: Applied Linear Algebra, Differential Equations, Mathematical Statistics I, Data Structures & OOP (Python), Introduction to Discrete Systems, Regression & ANOVA

Shelton High School — Top 2% · National AP Scholar w/ Distinction · RPI Medalist 2025

PROJECTS

Quantum Puzzle Solver, 1st Place, IBM Qiskit Fall Fest CT 2026 — Python, Qiskit, qBraid, AWS Braket Oct 2026
- Won UConn's Qiskit quantum hackathon (qBraid challenge) with a compact encoding that cut qubits from 24 to 12 and a constrained QAOA circuit that only produces valid states.
- Raised solve rate from 1.7% to 73.3% as circuit depth grew on AWS SV1 (0.15% random baseline); scaled to 24 qubits and verified every result against brute force.

Artist Streaming Analytics Pipeline — Python, PostgreSQL, pandas, Spotify for Artists, YouTube Data API
- Built an ETL pipeline merging Spotify for Artists exports and YouTube API data (500K+ streams) for 2 artists into PostgreSQL weekly.
- Wrote SQL models (CTEs, window functions) measuring release-day lift (avg +35%), playlist retention, and 30-day cohort decay.
- Turned findings into release-timing and ad-spend recommendations that contributed to a 25% rise in monthly listeners.

Volatility-Aware Trading System — Python, pandas, scikit-learn, Alpaca API, Datadog
- Built walk-forward backtesting over 5 years of daily data for 50 equities, cutting volatility forecast RMSE 12% vs a rolling baseline.
- Vol-targeted sizing: backtest Sharpe 1.05 vs 0.86 and max drawdown −14.1% vs −22.6% for equal-weight; runs live in a paper-trading bot.
- Deployed to a cloud instance with Datadog monitoring for order latency (210 ms median), fill rate (99%+), and live P&L.

AI Resume Screener — Claude API, MongoDB Atlas Vector Search, Python
- Parsed 500+ public-dataset resumes into structured documents, retrieved matches via embedding search, and scored role fit with an LLM.
- Reached 87% top-5 agreement with a 100-resume hand-labeled benchmark; tracked per-request cost (~$0.01) and latency (<4 s p95).

Real-Time Market Data Platform — Python, Redpanda (Kafka), TimescaleDB, FastAPI, Grafana, Docker In Progress
- Building a streaming pipeline that ingests live Alpaca equity trades and Kalshi order books through Redpanda into TimescaleDB, served via FastAPI endpoints and Grafana dashboards.

Kalshi Weather Prediction Model — Python, LightGBM, scikit-learn, NOAA NBM/MOS data, GitHub Actions In Progress
- Modeling P(daily high > threshold) for Kalshi temperature markets from NWS/NOAA forecasts with logistic regression and LightGBM (walk-forward), benchmarked by Brier score against market prices.

EXPERIENCE

Co-Founder & Lead Software Engineer | ClubIt LLC · Storrs, CT 2025 – Present
- Lead engineer and primary author of a multi-tenant SaaS platform for school club management (Next.js, TypeScript, Supabase), including a self-serve club registration flow for UConn.
- Found and fixed multi-tenant row-level security, role-escalation, and secret-management flaws across the Supabase backend.
- Built an outbound pipeline across 150+ CT and MA high schools (10+ demos booked); pursuing CAS and CREC procurement channels.

Technical Liaison, Interactive AI Theater | UConn Dept. of Dramatic Arts × CSE Senior Design Oct 2026 – Present
- Brought on by Prof. Sandi Carroll (Dramatic Arts) to translate a live AI-driven theater concept into technical goals for a CSE senior design team advised by Dr. Wei Wei.
- Proposed a scikit-learn approach that scores audience chat for quality, extracts themes instead of echoing messages, and trains on human-labeled aggregations.

Game Systems Developer (Luau/Lua) | Freelance · Remote 2020 – Present
- Engineered Luau gameplay systems (combat, UI, data persistence, monetization) for Roblox experiences with 1M+ combined visits.
- Optimized server- and client-side scripts, cutting server frame time ~30%; shipped features with designers via iterative workflows.

Marketing Intern & Artist Manager | Everlasting LLC · Remote Jan 2026 – Aug 2026
- Ran data-driven campaigns across Instagram, TikTok, and YouTube, growing artist followings 40% over 6 months.
- Managed an independent artist's career end-to-end, securing 10+ bookings and placements using engagement-driven positioning.

SKILLS

Languages: Python, SQL, TypeScript, Luau/Lua (6 yrs)
Data & ML: ETL pipelines, pandas, NumPy, scikit-learn, time-series analysis, regression & hypothesis testing, LLM APIs, vector search, quantum optimization (QAOA)
Tools: PostgreSQL, TimescaleDB, Supabase, MongoDB Atlas, Docker, Git/GitHub, GitHub Actions, Datadog, Grafana, Jupyter, Qiskit